

CredibleX Growth Intelligence Engine

Final Report

An independent, AI-assisted intelligence prototype connecting SME financing opportunities, product suitability, and partner prioritisation into one decision-support system for a CredibleX-style embedded-finance business.

Stack	Python · SQL · Excel · HTML / CSS / JavaScript · GenAI
Data	1,000 synthetic SMEs · 50 synthetic potential partners
Status	Independent portfolio prototype — not affiliated with CredibleX

1. Executive Summary

This project is a self-directed prototype exploring how a CredibleX-style embedded-finance lender could combine SME-level and partner-level intelligence into a single decision-support system. It was built end-to-end — synthetic data generation, a Python/SQL analytical engine, an AI explanation layer, and an interactive dashboard — to demonstrate the full pipeline a growth or data team would need, not just an isolated model.

Across the 1,000 simulated SMEs, the engine identified 59 as "High Opportunity" and a further 542 as "Emerging Opportunity." Revenue-Based Financing was the most frequently recommended product (419 SMEs), narrowly ahead of Receivables Financing (378) and Payable Financing (203). Of the 50 simulated potential partners, 15 were classified as "Priority," led by Supplier Network and Free Zone Authority partner types by average score.

The project's main design choice was to keep the AI layer strictly explanatory: every score is calculated deterministically in Python/SQL, and the AI is only ever prompted to narrate results it is handed — never to generate its own numbers (see Section 6).

2. Business Context

CredibleX is a UAE-based, FSRA-regulated SME lender that distributes three working-capital products — Revenue-Based Financing, Receivables Financing, and Payable Financing — primarily through embedded partnerships with platforms that already serve SMEs, rather than through direct customer acquisition. Public reporting describes a partner network that grew from roughly 35 to over 70 partners between 2024 and 2025–26, spanning payment infrastructure, free-zone authorities, FMCG distribution, and insurance (full detail in *docs/CredibleX_Research.md*).

That distribution-first model is the reason this project treats SME scoring and partner scoring as equally important, linked problems rather than treating partners as an afterthought to an SME credit model. The three real CredibleX products were used directly as the product-matching taxonomy, so the recommendation logic reflects an actual, current product suite rather than a generic one.

3. Methodology

The project follows a six-stage pipeline, each stage building on the last:

- **Data** — two synthetic datasets generated with realistic, correlated distributions (e.g. revenue scaled by industry, financing requirement driven by receivable days, growth, and cash buffer) rather than uniform random noise.
- **Cleaning & validation** — schema checks, duplicate-ID checks, and range/sanity checks run in `data_cleaning.ipynb` before anything downstream touches the data.
- **Analysis** — four analytical modules implemented in both Python (`opportunity_engine.ipynb`) and SQL (`analysis.sql`), so results are reproducible in either environment.
- **AI layer** — `ai_insights.ipynb` turns the calculated rows into structured business briefs and an executive summary, using prompts that only ever reference pre-computed fields (`ai/prompts.md`).
- **Dashboard** — a standalone HTML/CSS/JS application (no backend) renders the same scored data, and replicates the AI-brief template logic client-side so every one of the 1,000 SMEs and 50 partners — not just a sample — has a brief on click.
- **Reporting** — this document, summarising findings and reflecting on design trade-offs.

4. Analytical Framework

A. SME Financing Opportunity Score (0–100)

A weighted blend of five min-max normalised signals: revenue growth (25%), financing requirement relative to revenue (25%), working-capital pressure from receivable days and cash buffer (20%), transaction volume (15%), and revenue stability (15%). SMEs are then split into four segments using data-driven quantile thresholds (roughly top 6%, next 25%, next 40%, and the remainder) rather than arbitrary fixed cut-offs, so segment sizes stay meaningful regardless of how the underlying score distribution shifts.

B. Financing Product Recommendation

A rule-based matcher against CredibleX's actual three-product suite: Receivables Financing where receivable days are high relative to payable days; Revenue-Based Financing where growth is strong and revenue is stable; Payable Financing where payable days are high. Each recommendation carries a one-line, field-referencing reason.

C. Partner Opportunity Score (0–100)

Blends SME reach (28%), transaction volume (20%), financing relevance (27%), digital maturity (10%), and growth rate (15%), with a penalty subtracted for integration complexity. Segmented into Priority / Emerging / Monitor using the same quantile approach as the SME score.

D. Early Warning Classification

Deliberately kept simple: five binary flags (declining revenue, high receivable days, low transaction activity, low cash buffer, weak repayment history) are counted per SME. Three or more flags is "Attention Required," one or two is "Monitor," zero is "Stable." This is a screening signal for outreach sequencing, not a credit-risk model.

5. Key Findings

SME opportunity segmentation

Segment	SME Count	Share
High Opportunity	59	5.9%
Emerging Opportunity	542	54.2%
Moderate Opportunity	252	25.2%
Low Opportunity	147	14.7%

Financing product distribution

Product	SME Count
Revenue-Based Financing	419
Receivables Financing	378
Payable Financing	203

Construction (avg. score 44.2), Healthcare (41.9), and Trading (41.7) were the highest-scoring industries on average, though — reflecting how the synthetic financing-requirement signal was constructed — all ten industries clustered within a fairly narrow 5-point band, suggesting opportunity in this simulated portfolio is more a function of individual business characteristics than industry alone.

Total illustrative financing requirement across all 1,000 SMEs was approximately AED 194.7M; High-Opportunity SMEs alone carried an average individual requirement of roughly AED 356,000, notably above the portfolio mean.

On the partner side, 15 of 50 partners were classified Priority, 17 Emerging, and 18 Monitor. Supplier Network partners scored highest on average (53.9), followed by Free Zone Authority (47.5) and B2B Marketplace (44.9) partner types — broadly consistent with the partner categories CredibleX's real announcements emphasise (see Section 2).

Opportunity and risk are not the same axis. Cross-tabulating segment against early-warning status shows 6 of the 59 High-Opportunity SMEs (10%) are also flagged "Attention Required" — a reminder that a high commercial score does not automatically mean a safe financing candidate, and the two views need to be read together before outreach.

6. AI Layer

The AI layer's job is narrow and deliberate: explain pre-computed numbers in business language, never generate them. Every prompt (ai/prompts.md) injects the specific row's fields — industry, revenue, growth, receivable days, the already-calculated Opportunity Score and Recommended Product — and asks for a structured JSON response following a fixed six-part schema (Business Profile, Financing Opportunity, Recommended Product, Key Drivers, Potential Concern, Suggested Next Action for SMEs; an equivalent six-part schema for partners).

To keep the project runnable without API credentials — important for a portfolio piece a reviewer will actually execute — ai_insights.ipynb and the dashboard's script.js both implement a deterministic template

generator that follows the identical schema, and will automatically use a live Claude API call instead if an ANTHROPIC_API_KEY is present in the environment. This also made validation straightforward: because every sentence in a brief traces to a specific field, briefs can be spot-checked against the source row for correctness.

A separate, simpler AI Executive Summary synthesises four portfolio-level insights (Portfolio, Product, Partner, Early Warning) from aggregate statistics rather than per-row data — these four insights are what the dashboard's "AI Insights" page displays.

7. Challenges & Solutions

Challenge	Solution
No internal CredibleX data	Synthetic dataset with realistic, correlated distributions, grounded in public research on CredibleX's actual product suite and partner categories.
Generic credit-model risk	Scores explicitly framed as a commercial-opportunity signal, not credit risk; early-warning kept as a separate, simple, transparent flag count rather than a black-box risk model.
AI hallucination	AI restricted to explaining pre-computed, injected fields via structured prompts; template fallback guarantees identical structure without live API access.
Complex analysis	Four modules kept intentionally simple (weighted normalisation, rule-based matching, flag counting) and mirrored exactly between Python and SQL rather than layering in a black-box ML model.
1,000+ records needing individual briefs	AI-brief template logic reimplemented client-side in the dashboard, so every SME and partner — not just a sample exported to JSON — has a brief available on click.

8. Business Impact

While built on synthetic data, the pipeline demonstrates a pattern that could plausibly support several real workflows for an embedded-finance lender:

- **SME opportunity discovery** — surfacing the highest-opportunity SMEs from a large base for proactive outreach, rather than waiting for inbound applications.
- **Product targeting** — pairing outreach with a specific, justified product recommendation instead of a generic financing pitch.
- **Partner prioritisation** — focusing embedded-finance business-development effort on the partners most likely to convert reach into financing volume.
- **Business-development research** — AI Partner Briefs give a BD team a ready-made starting point (fit, angle, growth trajectory) for a first outreach conversation.
- **Portfolio monitoring** — the early-warning layer offers a lightweight way to flag SMEs needing a closer look before financing offers go out, independent of the opportunity score.

9. Limitations & Future Scope

The central limitation is that both datasets are synthetic: distributions were designed to be realistic and internally consistent, but they are not real CredibleX SME or partner data, and the specific figures in Section 5 should be read as illustrative of the method, not as claims about CredibleX's actual portfolio. Similarly, without access to CredibleX's real underwriting methodology, the scoring weights in Section 4 are reasoned assumptions rather than calibrated, validated weights — a production version would need to backtest and tune these against real outcomes.

Future extensions could include: calibrating score weights against real repayment/conversion outcomes; adding a time dimension so opportunity and early-warning scores update as SME behaviour changes month to month; extending the partner model with actual integration cost/effort data; and replacing the template AI fallback with a fully live LLM pipeline, with human review before any brief reaches an SME- or partner-facing team.

10. Conclusion

This prototype set out to show that an embedded-finance lender's two core growth questions — which SMEs need financing, and which partners can reach them — are best answered together, not separately. By building a small, transparent analytical core in Python and SQL, wrapping it in an AI layer that explains rather than invents, and surfacing all of it through a dashboard that runs standalone from a single link, the project demonstrates a complete, reproducible pattern for AI-assisted growth intelligence — one grounded in CredibleX's real, publicly described business model, even though the underlying data here is synthetic.

Independent prototype built for portfolio purposes. Not affiliated with, commissioned by, or endorsed by CredibleX. All figures in this report derive from synthetically generated data.